

Rancang Bangun Sistem Monitoring Risiko Kebakaran Berbasis .NET MAUI dengan Integrasi Face Recognition, Gemini AI, Fuzzy Logic, Neural Network, dan SQLite Database

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Abstract—Fire risk in indoor and public environments requires fast monitoring, early warning, and understandable recommendations for both operators and non-technical users. Conventional fire monitoring systems generally display sensor values without intelligent interpretation, while user authentication and historical data storage are often not integrated into a single cross-platform application.

This research designs and implements a .NET MAUI-based fire risk monitoring system integrating simulated temperature, smoke, and humidity readings, SQLite Database, Face Recognition, Fuzzy Logic, Artificial Neural Network (ANN), and a Natural Language Processing (NLP) assistant using the Gemini API. Temperature, smoke concentration, and humidity are processed in real time to generate a danger score and status classification. Fuzzy Logic is used to convert uncertain sensor conditions into linguistic outputs, while ANN performs feedforward prediction and backpropagation-based training on representative fire-risk patterns. Face Recognition using Haar

I. INTRODUCTION

Fire hazards in buildings, laboratories, residential areas, and public facilities can develop rapidly when increases in temperature and smoke are not detected at an early stage. A monitoring system is therefore required not only to display environmental values but also to provide clear warnings and recommendations for people who do not understand technical sensor readings.

Modern cross-platform application development enables a single codebase to be deployed on multiple devices. .NET MAUI provides a practical framework for building user interfaces, authentication pages, dashboards, and local database operations within one application. In this project, .NET MAUI is used to implement a fire risk monitoring application with two user roles: Operator and Warga Sipil.

The proposed system integrates several intelligent components. Face Recognition is used to strengthen login security, SQLite stores user accounts and monitoring history, Fuzzy Logic interprets uncertain sensor values into human-readable fire risk page, operator dashboard, citizen dashboard, and AI assistant panel. SQLite is used as a local database to store user accounts, sensor readings, chat logs, and broadcast messages, enabling the system to keep history even when the application is used offline.

C. Face Recognition

Face Recognition is a biometric authentication technique that verifies the identity of a user based on facial characteristics. This project applies OpenCV Haar Cascade for face detection and LBPH Face Recognizer for comparing a live camera image with a stored registration image. A threshold-based mechanism and several consecutive successful frames are used to reduce accidental verification.

D. Fuzzy Logic

Fuzzy Logic is suitable for monitoring problems involving uncertainty and gradual transitions between conditions. Instead of producing only binary decisions, Fuzzy Logic uses membership functions and inference rules to classify conditions such as safe, warning, and danger. This approach is useful for

Cascade and LBPH is applied as a second authentication layer after username and password verification.

The developed system provides two role-based dashboards: an Operator dashboard with detailed monitoring, AI analysis, Gemini chatbot, database test command, and broadcast messaging; and a Warga Sipil dashboard with simplified safety information and operator messages. Test scenarios show that the system can perform registration, database-based login, live face verification on Windows, real-time monitoring, local Fuzzy-ANN interpretation, Gemini-based explanation, and SQLite historical storage. The system demonstrates the feasibility of combining cross-platform UI, AI reasoning, and local database storage in a fire monitoring application that can be understood by ordinary users.

Index Terms—Fire Risk Monitoring, .NET MAUI, Fuzzy Logic, Artificial Neural Network, Face Recognition, Gemini API, NLP, SQLite Database.

categories, ANN predicts risk patterns through feedforward and backpropagation mechanisms, and an NLP assistant based on the Gemini API provides conversational explanations and recommendations. This integration is intended to make the monitoring result easier to understand and more useful for decision making.

II. LITERATURE REVIEW

A. Fire Risk Monitoring

Fire risk monitoring is the process of observing physical indicators such as temperature, smoke, and humidity to determine whether an area is safe, warning-level, or dangerous. Temperature and smoke are dominant indicators of combustion, while low humidity can increase the likelihood of fire spread. In a decision-support system, these parameters can be combined into a single risk index for easier interpretation.

B. .NET MAUI and Local Data Storage

.NET MAUI enables the development of cross-platform applications using C# and XAML. In this system, .NET MAUI manages the login interface, registration form, face verification interpreting fire risk because temperature, smoke, and humidity may not always produce clear boundaries.

E. Artificial Neural Network

Artificial Neural Network (ANN) is a computational model inspired by biological neurons. ANN learns relationships between input data and target outputs through weight adjustment. In this system, the ANN receives temperature, smoke, and humidity as input neurons, processes them through a hidden layer, and produces a predicted fire risk value through an output neuron. Backpropagation is used to update weights during training.

F. Natural Language Processing

Natural Language Processing allows users to ask questions using ordinary language. This system combines the Gemini API as a cloud-based NLP engine with a local rule-based fallback. The Gemini API processes contextual questions from the operator, while local NLP handles common commands such as status checking, database testing, fuzzy analysis, neural network analysis, and broadcast messaging.

III. RESEARCH METHODOLOGY

A. System Architecture

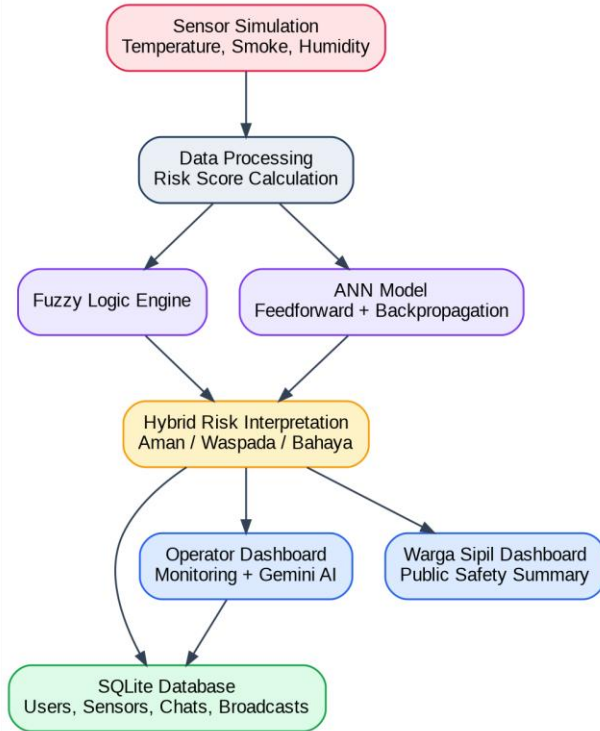


Figure 1. Proposed FIRE system architecture.

The proposed architecture starts from environmental data acquisition. In this prototype, temperature, smoke, and humidity values are simulated inside the dashboard to validate the logic flow before physical sensors are integrated. The values are processed by the risk calculation module, Fuzzy Logic engine, and ANN model. The results are displayed in role-based dashboards, stored in SQLite, and used as context by the Gemini AI assistant.

B. UML and System Flowchart

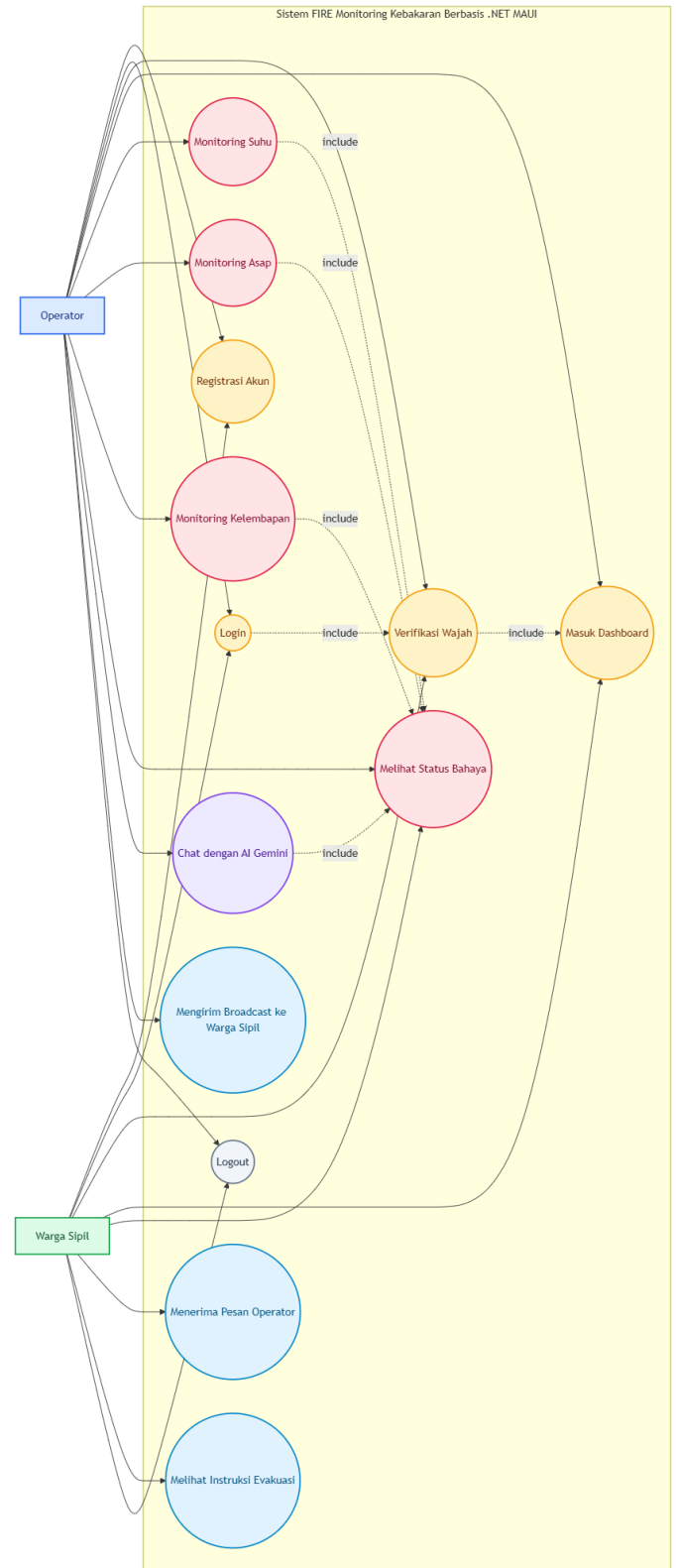


Figure 2. Overall system flowchart.

The system is modeled using use case, class, sequence, and activity diagrams. The main actors are Operator and Warga Sipil. The Operator can monitor sensor values, access detailed AI analysis, ask the chatbot, and broadcast messages. Warga Sipil can view simplified safety status, AI summary, and operator messages. The main classes include MainPage, RegisterPage, FaceVerificationPage, MainDashboardPage, GeminiAiService, FireDatabase, FireFuzzyService, and FireNeuralNetworkService.

C. Database Design

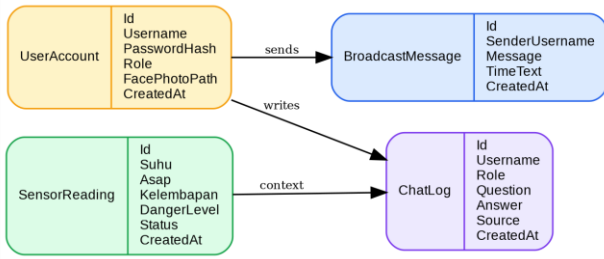


Figure 3. SQLite database schema used in the system.

The database design consists of four main entities: UserAccount for account and role information, SensorReading for periodic environmental records, ChatLog for storing questions and answers from the AI assistant, and BroadcastMessage for storing operator messages. User passwords are stored as hashes instead of plain text.

D. Fuzzy-Neural Decision Method

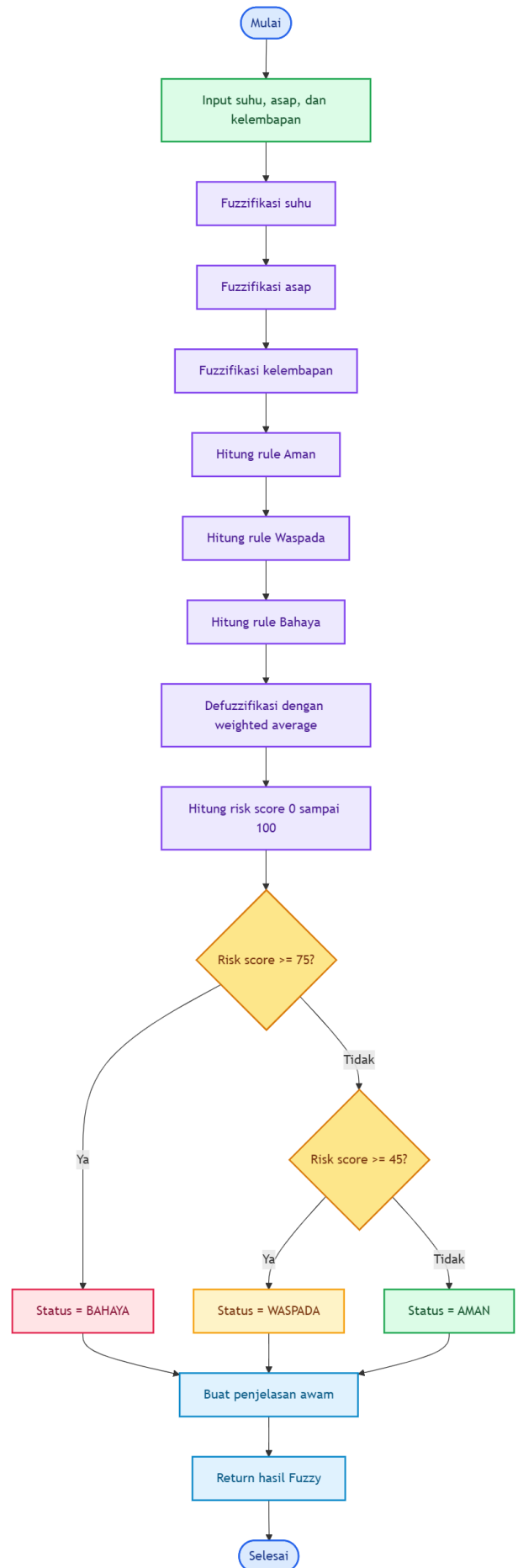


Figure 4. Fuzzy Logic processing flow.

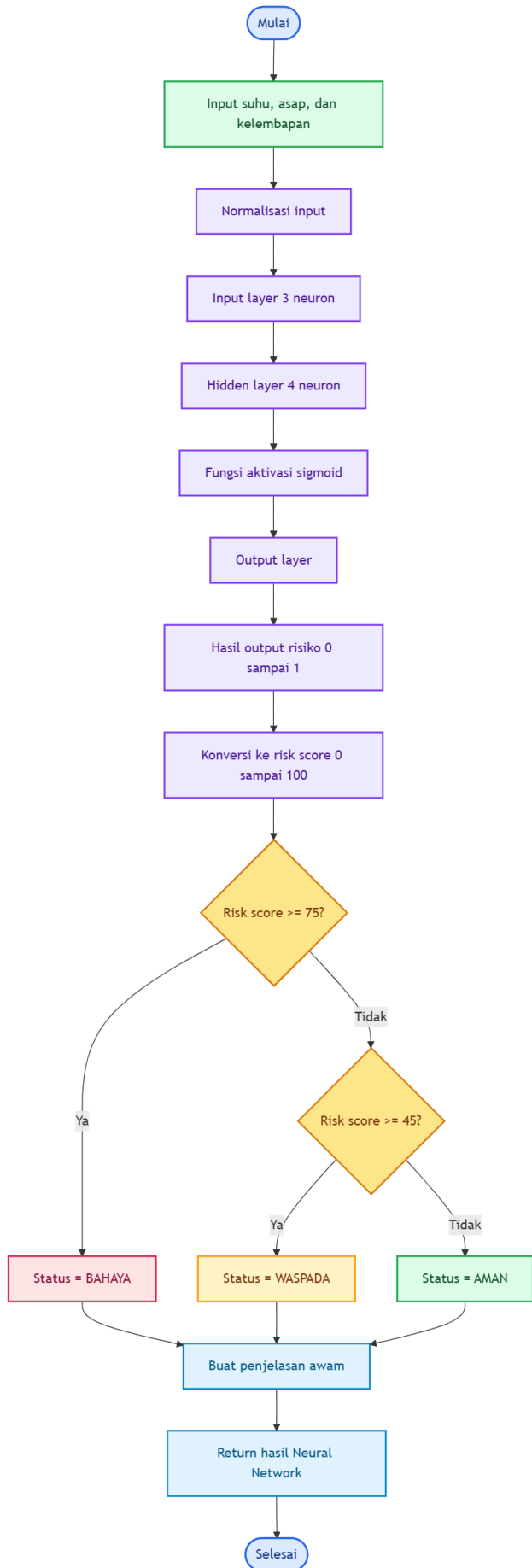


Figure 5. Neural Network feedforward and backpropagation flow.

The Fuzzy Logic module applies membership functions for temperature, smoke, and humidity. The inference rules produce three degrees of output: Aman, Waspada, and Bahaya. The final Fuzzy risk score is calculated using a weighted average. The ANN module normalizes sensor inputs, calculates hidden-layer activations using sigmoid functions, generates an output risk value, and applies backpropagation during training. The final dashboard shows both Fuzzy and Neural Network results to give a clearer explanation for users.

TABLE I ARTIFICIAL NEURAL NETWORK CONFIGURATION PARAMETERS

Parameter	Value
Input layer	3 neurons: T, smoke, RH
Hidden layer	4 neurons
Output layer	1 risk neuron
Activation	Sigmoid
Training	FF + Backprop
Epochs	3000
Learning rate	0.08

TABLE II ARTIFICIAL NEURAL NETWORK TRAINING DATASET

Temp (C)	Smoke (ppm)	Humidity (%)	Target	Status
25	100	80	0.05	Aman
30	200	60	0.15	Aman
40	300	45	0.35	Waspada
50	450	45	0.5	Waspada
55	500	40	0.6	Waspada
65	650	35	0.75	Bahaya
75	800	30	0.95	Bahaya
85	900	25	0.99	Bahaya

IV. SYSTEM IMPLEMENTATION

A. Login and Registration Implementation

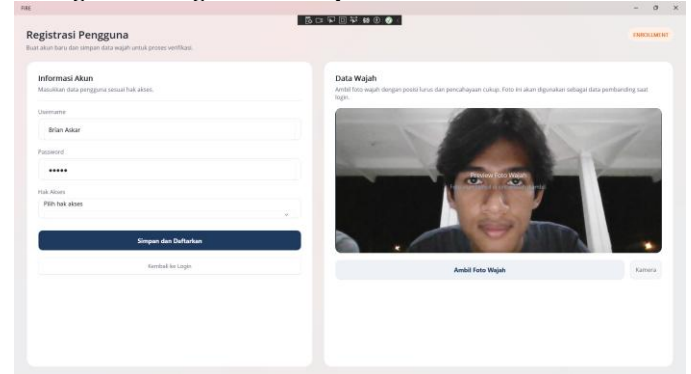


Figure 6. Face Verification Process Using Haar Cascade

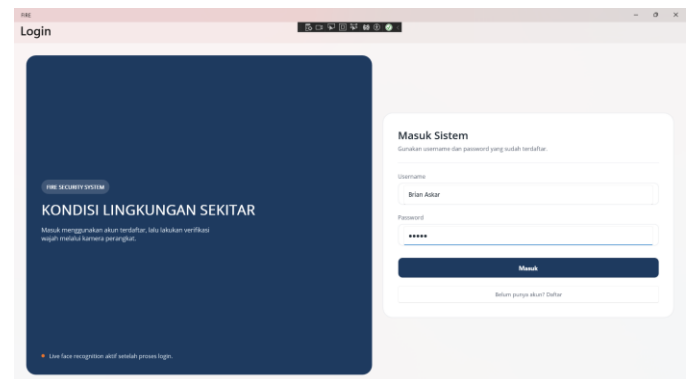


Figure 7. System Login Screen

The login and registration modules are implemented using MainPage and RegisterPage. A new user registers by entering username, password, role, and facial image. The photo is saved in the application data directory, while the account information is inserted into the SQLite database. During login, the application retrieves the user record from the database and verifies the password hash before continuing to the face verification page.

B. Face Recognition Implementation

Face verification is implemented using camera-based scanning on Windows. The system loads the registered face image, detects the face region using Haar Cascade, normalizes the image, and trains an LBPH recognizer using the reference face and its flipped version. Live frames are compared with the trained recognizer. Verification is accepted only when the confidence value is within the threshold for several consecutive frames.

TABLE III FACE RECOGNITION CONFIGURATION AND TEST RESULT

Parameter	Description
Detection	Haar Cascade
Recognizer	LBPH
Threshold	60.0
Success frames	5
Platform	Windows
Status	Valid if matched

C. Operator Dashboard

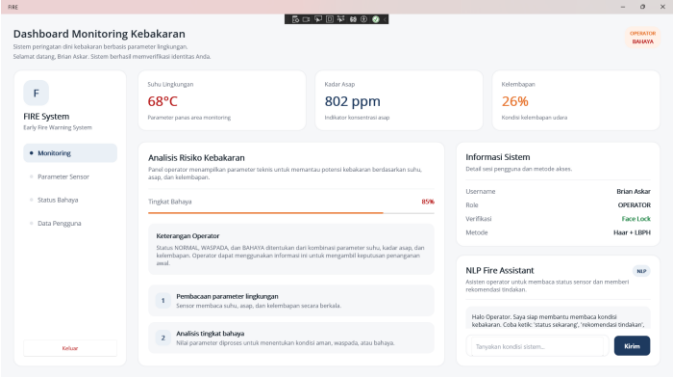


Figure 8. Operator Dashboard

The Operator dashboard displays real-time temperature, smoke, humidity, danger percentage, system status, detailed Fuzzy Logic output, Neural Network prediction, Gemini chatbot, database test command, and broadcast messaging. The Operator can ask questions such as status, recommendations, database condition, fuzzy analysis, and neural network analysis. The dashboard also saves sensor records to SQLite every ten seconds.

D. Warga Sipil Dashboard

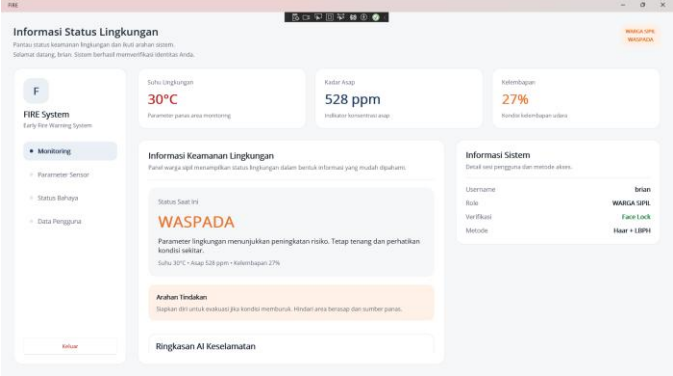


Figure 9. Warga Sipil Dashboard

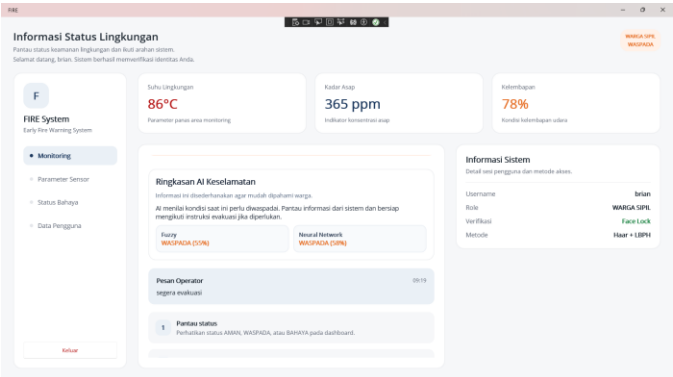


Figure 9. Warga Sipil Dashboard

The Warga Sipil dashboard presents simplified safety information. Instead of showing detailed algorithmic outputs only, it provides public status, instructions, operator messages, and a concise AI safety summary. This role-based simplification is intended to make the system understandable for ordinary users without technical background.

E. Gemini AI Assistant and NLP Fallback

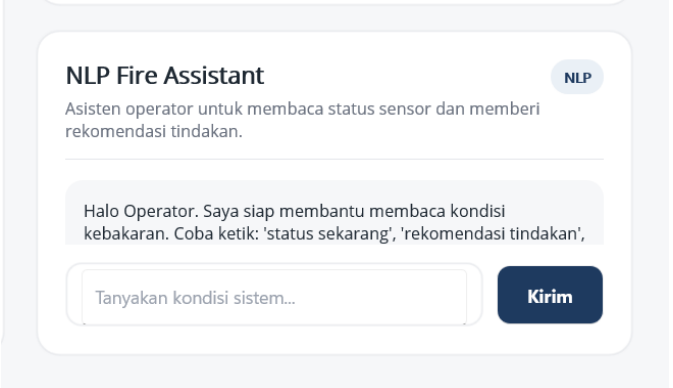


Figure 10. Gemini AI Assistant

The AI assistant uses the Gemini API through GeminiAiService. The prompt includes the latest temperature, smoke, humidity, status, and danger level. To reduce token usage, maxOutputTokens is limited to 350 and a lightweight Gemini model is selected. If the API fails, the system uses a local rule-based NLP response generator so that the dashboard remains functional.

F. Fuzzy Logic and Neural Network Display

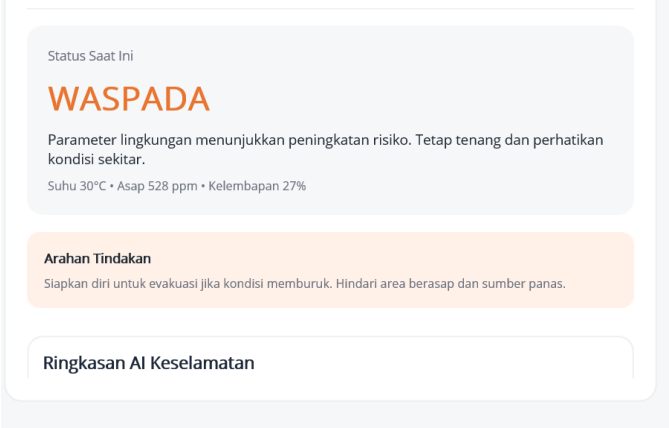


Figure 11. Warga Sipil Dashboard

Fuzzy and Neural Network analysis can be accessed through chatbot commands and are also displayed automatically on both dashboards. This design ensures that users do not need to manually request AI analysis. Each sensor update triggers UpdateAiAnalysisPanels, which recalculates Fuzzy and ANN predictions and updates the corresponding labels on the dashboard.

V. RESULTS AND DISCUSSION

A. Sensor Simulation and Risk Score Testing

Testing was performed using simulated values generated in the dashboard. These values were used to verify whether the danger calculation, status display, Fuzzy output, ANN output, database storage, and AI assistant response behaved consistently under different conditions.

TABLE IV SENSOR SIMULATION TEST RESULTS

Temp (C)	Smoke (ppm)	Humidity (%)	Danger (%)	Status
28	120	75	0	NORMAL
38	260	55	20	NORMAL
52	450	44	60	WASPADA
66	720	32	85	BAHAYA
78	850	28	100	BAHAYA

B. Fuzzy Logic Testing

Fuzzy Logic testing shows that the system produces gradual results rather than abrupt binary classification. When smoke and temperature increase while humidity decreases, the Fuzzy risk score rises and the status changes from Aman to Waspada and then Bahaya.

TABLE V FUZZY LOGIC TEST RESULTS

Temp	Smoke	Hum	Fuzzy Index	Status
25	100	80	20	AMAN
40	300	45	43	AWAS
55	500	40	58	WASPADA
65	700	35	76	BAHAYA
80	900	25	90	BAHAYA

C. Artificial Neural Network Testing

The ANN was trained using representative samples of fire risk conditions. The decreasing error value indicates that the ANN learns the relationship between temperature, smoke, humidity, and target risk level. The model is not intended as a final industrial-grade predictor but demonstrates feedforward and backpropagation integration inside the .NET MAUI application.

TABLE VI ANN TRAINING CONVERGENCE

Epoch	Mean Squared Error
500	0.082
1000	0.054
2000	0.031
3000	0.018

D. Hybrid Interpretation Testing

The system displays both Fuzzy Logic and ANN results. Instead of hiding algorithmic differences, the dashboard presents each output to the operator and summarizes them for Warga Sipil. The combined interpretation provides a more stable explanation because Fuzzy Logic offers transparent rules while ANN provides pattern-based estimation.

TABLE VII FUZZY-NEURAL DECISION TEST RESULTS

Sample	Fuzzy	ANN	Average	Status
1	20	18	19	AMAN
2	43	46	44.5	AWAS
3	58	62	60	WASPADA
4	76	79	77.5	BAHAYA
5	90	93	91.5	BAHAYA

E. Database and AI Assistant Testing

Database testing was performed by registering a user, logging in, entering the dashboard, waiting for sensor records to be saved, asking the chatbot, and sending an operator broadcast. The database test command returns the SQLite path and counts of stored sensor and chat logs. The AI assistant was tested using questions such as "kamu siapa", "buat laporan singkat", "rekomendasi tindakan", "analisis fuzzy", and "analisis neural network".

TABLE VIII OVERALL SYSTEM MODULE TEST SUMMARY

Module	Test Result
Register/Login	SQLite user data
Face verify	Windows target
Monitoring	2-second update
SQLite	10-second sensor log
Gemini AI	Contextual answers
Fuzzy	Risk score + status
ANN	Predicted risk
Broadcast	Operator to warga

F. Overall Discussion

The results show that the system successfully integrates several modules into a single .NET MAUI application. The Operator dashboard provides detailed technical information, while the Warga Sipil dashboard presents simplified safety guidance. This role-based design supports both technical monitoring and public communication. The main limitation is that the environmental data is still simulated; future development should integrate physical sensors such as temperature sensors, smoke sensors, and humidity sensors for real-time field measurement.

VI. CONCLUSION

This research successfully designed and implemented a .NET MAUI-based fire risk monitoring system integrating SQLite Database, Face Recognition, Fuzzy Logic, Artificial Neural Network, Gemini API, and local NLP fallback. The system supports user registration, database-based login, biometric verification, role-based dashboard presentation, AI-assisted explanation, and historical data storage.

The Fuzzy Logic module translates uncertain environmental conditions into interpretable risk categories, while the ANN module demonstrates feedforward and backpropagation for pattern-based fire risk prediction. Displaying both methods directly on the Operator and Warga Sipil dashboards improves transparency and makes the information easier to understand for non-technical users.

Future work should focus on integrating physical IoT sensors, improving face recognition robustness using deep learning, synchronizing SQLite data with a cloud database, and adding notification mechanisms for mobile devices so that warnings can be delivered faster during emergency conditions.

CONTRIBUTION OF THIS RESEARCH

1. Development of a cross-platform fire monitoring application using .NET MAUI.
2. Implementation of SQLite-based account, sensor, chat, and broadcast storage.
3. Integration of Face Recognition authentication using OpenCV Haar Cascade and LBPH.
4. Integration of Gemini API and local rule-based NLP fallback for AI assistant features.
5. Implementation of Fuzzy Logic for interpretable fire risk classification.
6. Implementation of ANN feedforward and backpropagation for risk prediction.
7. Presentation of AI analysis directly on role-based dashboards for Operator and Warga Sipil.

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